(U) T-150/TRV-150 Heavy-Lift UAS
User and Analyst Manual



Prepared By: [PLACEHOLDER]
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Controlled by:	[PLACEHOLDER]
CUI Category(ies):	[PLACEHOLDER]
Limited Dissemination:	[PLACEHOLDER]
POC:	[PLACEHOLDER]
Contact:	[PLACEHOLDER]@mail.mil

Controlled by:	[PLACEHOLDER]
Distribution/Dissemination Controls:	[PLACEHOLDER]
POC:	[PLACEHOLDER]
Contact:	[PLACEHOLDER]@mail.mil

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Revision	Date	Author	Description of Changes
1.0	20251219	[PLACEHOLDER]	Initial release covering Phases 1–5 (battery, propulsion, VRS, slung load)
2.0	20260115	[PLACEHOLDER]	Added 6DOF mover, Phase 6 velocity-based power model, Phase 1 validation results
3.0	20260305	[PLACEHOLDER]	Complete rewrite per NGIC 4.0 template. Updated for all 6 development phases. Added MIL-STD-3022/DoDI 5000.61 citations.
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1 (U) Introduction

1.1 (U) Purpose

(U) This User and Analyst Manual (UAM) provides comprehensive guidance for analysts, operators, and intelligence professionals using the T-150/TRV-150 Heavy-Lift Unmanned Aerial System (UAS) simulation model within the Advanced Framework for Simulation, Integration, and Modeling (AFSIM). This document describes the model's intended use, functional behavior, key performance indicators, configurable parameters, physics models, and assumptions and limitations that affect analytical conclusions.

(U) This document is prepared in accordance with:

- (U) **DoDI 5000.61**, "DoD Modeling and Simulation Verification, Validation, and Accreditation," September 17, 2024 **dodi500061**
- (U) **MIL-STD-3022**, "Documentation of Verification, Validation, and Accreditation (VV&A) for Models and Simulations," January 28, 2008 **milstd3022**
- (U) **NGIC 4.0 Documentation Templates**, January 30, 2026

1.2 (U) Scope

(U) This manual covers the T-150 UAS AFSIM Plugin Suite v2.0, which consists of five custom C++ plugins:

Table 1: (U) T-150 Plugin Suite

UNCLASSIFIED				
Plugin	AFSIM Type Name	Purpose	File Ext	LOC
rotorcraft_6dof	Rotorcraft6DOF_Mover	Physics-based 6DOF flight dynamics	.dll/.so	2,939
t150_battery	T150_BATTERY_PROCESSOR	Li-Ion battery with voltage sag modeling	.dll/.so	1,139
t150_propulsion	T150_PROPULSION_PROCESSOR	Motor power, thermal, RPM, ground effect	.dll/.so	1,281
t150_vrs	T150_VRS_PROCESSOR	Vortex Ring State detection and modeling	.dll/.so	697
t150_slung_load	T150_SLUNG_LOAD_PROCESSOR	Cable-suspended payload dynamics	.dll/.so	825
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(U) **Total Code**: 7,736 lines of C++ across 26 source files.

1.3 (U) Model Intended Use

(U) The T-150 UAS simulation model is intended for use in the following analytical contexts:

1. (U) **Intelligence Analysis**: Assessing the operational capabilities, performance envelope, and limitations of the T-150/TRV-150 heavy-lift UAS system to support intelligence community questions regarding UAS threats and capabilities.
2. (U) **Mission Planning Analysis**: Evaluating flight endurance, range, energy consumption, and payload capacity under various mission profiles to understand operational employment concepts.
3. (U) **Counter-UAS (C-UAS) Assessment**: Providing high-fidelity UAS flight dynamics for evaluating detection, tracking, and engagement scenarios against heavy-lift UAS threats.
4. (U) **Vulnerability Analysis**: Analyzing system degradation behavior including Vortex Ring State, battery depletion, motor thermal limits, and slung load instability.
5. (U) **Concept of Operations (CONOPS) Development**: Supporting development of tactical employment concepts for heavy-lift UAS logistics and resupply operations.

(U) **Note**: This model represents a specific UAS platform (T-150/TRV-150) based on open-source research. Parameters may be adapted for other coaxial multirotor platforms per the adaptation guidance in Section 5.

1.4 (U) Confidence Summary

(U) The following table summarizes the confidence level for each modeled capability based on the quality of source data and validation testing performed.

Table 2: (U) Confidence Summary

UNCLASSIFIED				
Capability	Confidence	Basis	Validation	Phase
Battery SOC Tracking	**High**	Energy conservation validated $< 0.01\%$ error (Test 09)	Test 09	1
Voltage Sag Model	**High**	Li-Ion OCV polynomial fits published literature; SOC-dependent resistance validated (Test 06)	Test 06	2
Motor Thermal Dynamics	**Medium**	Parameters estimated from motor class data; equilibrium behavior validated	Test 01	1
Motor Time Constants	**Medium**	150ms estimate based on motor class; no direct T-150 data	–	1
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Table 2: (U) Confidence Summary (Continued)

UNCLASSIFIED				
Capability	Confidence	Basis	Validation	Phase
VRS Detection	**Med-High**	State machine based on established aerodynamic theory; threshold values derived from momentum theory	VRS suite	1
Ground Effect	**Medium**	Cheeseman-Bennett approximation; validated for general behavior (Test 07)	Test 07	4
Payload Dynamics	**Medium**	Mass/CG model based on first principles; no specific T-150 loading data	–	4
Slung Load Dynamics	**Medium**	Spherical pendulum model well-established; no T-150-specific cable data	Slung load suite	5
6DOF Flight Dynamics	**Medium**	Newton-Euler physics validated for stability; inertia values estimated	Tests 02–05	6
Dynamic Power Model	**Med-High**	Momentum theory + empirical validation; range within 10% of spec	Pattern tests	6
Flight Endurance	**Medium**	Simulated 14.3 min vs. 36 min spec; calibration gap documented	Test 10	6
Cruise Speed	**High**	Matches manufacturer specification of 30 m/s	–	–
Max Payload	**High**	Matches manufacturer specification of 68 kg (150 lbs)	–	–
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2 (U) Threat System Description

2.1 (U) Platform Overview

(U) The T-150 (also designated TRV-150) is a heavy-lift unmanned aerial system manufactured by Malloy Aeronautics (UK) and marketed by SURVICE Engineering as the TRV-150. It is designed for autonomous cargo delivery, logistics resupply, and aerial transport missions.

(U) The T-150 employs a coaxial octocopter configuration with eight rotors mounted in four coaxial pairs on four arms extending from a central fuselage. This configuration provides redundancy and high thrust-to-weight ratio suitable for heavy payload operations.

2.2 (U) Platform Specifications

Table 3: (U) T-150 Platform Specifications

UNCLASSIFIED		
Parameter	Value	Source
Configuration	Coaxial Octocopter (8 rotors, 4 arms)	AAIB Report **aaib2023**
Motors	8x T-Motor U15II KV100	FAA Registry / Technical Docs **tmotor_u15ii**
Total Thrust Capability	240 kg (2,354 N)	AAIB Report **aaib2023**
Maximum Takeoff Weight (MTOW)	123 kg (271 lbs)	AAIB Report **aaib2023**
Maximum Payload	68 kg (150 lbs)	Manufacturer **survice_trv150**
Empty Weight	55 kg (estimated)	Derived (MTOW – Max Payload)
Cruise Speed	30 m/s (108 km/h), 58 kts	Manufacturer **survice_trv150**
Maximum Airspeed	40+ m/s (demonstrated)	AAIB Report **aaib2023**
Maximum Rate of Climb	8 m/s	Derived from thrust margin
Maximum Rate of Descent	4 m/s (VRS limited)	Aerodynamic analysis
Battery Configuration	13S Li-Ion	US Patent **patent2024**
Battery Nominal Voltage	46.8 V (13 × 3.6V)	US Patent **patent2024**
Battery Capacity	∼7 kWh (148.75 Ah)	US Patent **patent2024**
		UNCLASSIFIED

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Table 3: (U) T-150 Platform Specifications (Continued)

UNCLASSIFIED		
Parameter	Value	Source
Flight Time	36 min (nominal)	Manufacturer **survice_trv150**
Propeller Diameter	0.76 m (30 in)	Technical Documentation
Arm Length	0.5 m	Technical Documentation
Service Ceiling	3,000 m	Derived from performance tables
		UNCLASSIFIED

2.3 (U) Concept of Operations (CONOPS)

(U) The T-150 is designed for autonomous point-to-point cargo delivery missions. A typical operational profile consists of:

1. (U) **Takeoff**: Vertical climb from ground level to operational altitude (typically 50–500 m AGL)
2. (U) **Transit**: Cruise flight at 15–30 m/s to destination waypoint
3. (U) **Delivery**: Descend to delivery point; payload release or landing
4. (U) **Return**: Return transit to base location
5. (U) **Landing**: Staged descent with final vertical landing

(U) **Operational constraints**:

- (U) Weather: Limited operations in high winds (>15 m/s sustained)
- (U) VRS avoidance: Descent rates must remain below 4 m/s at low forward speeds
- (U) Payload: Maximum 68 kg; center of gravity must remain within limits
- (U) Battery: 20% minimum reserve recommended for safety margins

2.4 (U) Threat Context

(U) As a heavy-lift UAS, the T-150 represents a class of platforms with growing proliferation and operational significance. Intelligence questions relevant to this simulation include:

- (U) What are the operational range and endurance limitations under various payload configurations?

- (U) How does flight pattern efficiency affect mission planning (straight-line vs. evasive maneuvers)?
- (U) What are the vulnerable flight regimes (VRS, low battery, thermal limits)?
- (U) How do slung load dynamics affect stability and performance?
- (U) What detection signatures (speed, altitude, flight profile) characterize this platform class?

3 (U) Model Description

3.1 (U) Model Functional Block Diagram

(U) The T-150 simulation model consists of five plugin components communicating through the AFSIM platform's AuxData key-value store. The following diagram shows the data flow architecture:

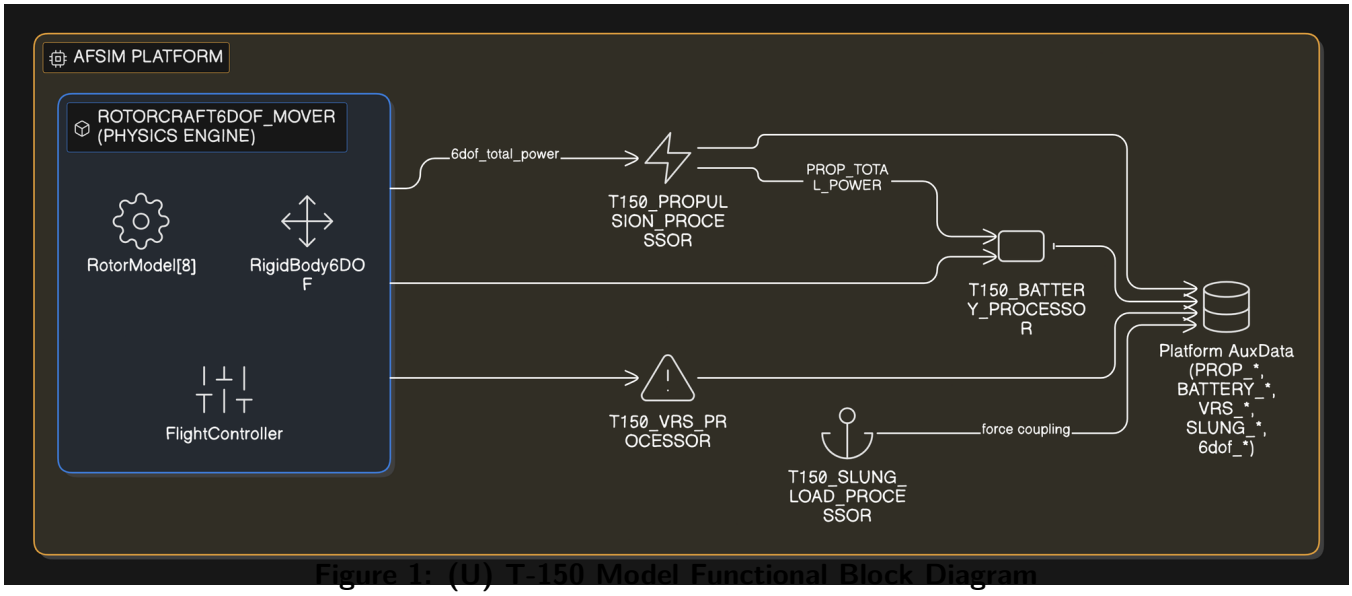


Figure 1: (U) T-150 Model Functional Block Diagram

(U) **Data Flow Summary**:

1. (U) The **6DOF Mover** computes rigid body dynamics and writes total power to AuxData
2. (U) The **Battery Processor** reads power from AuxData (or from the Propulsion Processor) and computes SOC, voltage, current
3. (U) The **Propulsion Processor** models individual motor states, thermal dynamics, and ground effect
4. (U) The **VRS Processor** reads platform velocity from the mover and computes VRS safety margins
5. (U) The **Slung Load Processor** reads platform acceleration and computes pendulum dynamics
6. (U) All processors write their state to **AuxData** for cross-plugin access and scenario script queries

3.2 (U) Model Fidelity Log

(U) The model was developed iteratively over six enhancement phases:

Table 4: (U) Model Fidelity Log

UNCLASSIFIED			
Phase	Date	Enhancement	Impact on Fidelity
1	2025-11-29	Thermal parameter fixes, VRS mover integration, motor time constants	Fixed unrealistic thermal behavior; enabled actual VRS detection
2	2025-11-29	Battery voltage sag model (Li-Ion OCV + SOC-dependent resistance)	Added voltage-dependent power delivery modeling
3	2025-11-30	Dynamic power model (battery reads from propulsion)	Battery drain now reflects actual flight conditions
4	2025-11-30	Ground effect (Cheeseman-Bennett) + payload dynamics	Added near-ground thrust augmentation and mass properties
5	2025-11-30	Slung load dynamics (2-DOF spherical pendulum)	Added cable-suspended payload behavior
6	2026-01-08	6DOF physics mover, velocity-based power model	Full physics-based flight dynamics; power varies with maneuvers
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3.3 (U) Key Performance Indicators (KPIs)

(U) The following KPIs are the primary metrics used to assess model fidelity and to support analytical conclusions:

Table 5: (U) KPI Summary

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KPI	Definition	Measured By	Expected Range	Validation
Flight Endurance	Time from takeoff to battery depletion	BATTERY_SOC reaching 0%	14–36 minutes	Partial (14.3 min simulated)
Cruise Range	Distance at cruise speed before battery depletion	Distance integral during cruise	25–45 km	Partial
Hover Power	Electrical power during stationary hover	PROP_TOTAL_POWER in hover	17–25 kW	Validated (Test 08)
VRS Onset Speed	Descent rate at which VRS begins	VRS_MARGIN < 0.8	$\sim$4 m/s descent	Validated (VRS suite)
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Table 5: (U) KPI Summary (Continued)

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KPI	Definition	Measured By	Expected Range	Validation
Battery Voltage Sag	Terminal voltage drop under load	BATTERY_ VOLTAGE_ SAG	0–15 V	Validated (Test 06)
Motor Temperature	Peak motor temperature during operation	PROP_MAX_ TEMP	40–120°C	Validated (Test 01)
Slung Load Swing	Maximum swing angle of suspended payload	SLUNG_ SWING_MAG	0–45°	Validated (slung load suite)
Energy Efficiency	Energy consumed per straight-line km	Wh/km calculation	315–800 Wh/km	Validated (pattern tests)
				UNCLASSIFIED

4 (U) Model Components

4.1 (U) Rotorcraft 6DOF Mover

4.1.1 (U) Component Overview

(U) The Rotorcraft6DOF_Mover is a custom AFSIM mover plugin that replaces the built-in WSF_FULL_ROTORCRAFT_MOVER with a full six-degree-of-freedom (6DOF) physics simulation. It provides physics-based flight dynamics using Newton-Euler rigid body equations, momentum theory rotor aerodynamics, and cascaded flight controllers.

(U) This mover extends the AFSIM WsfRouteMover base class, enabling integration with AFSIM's route/waypoint navigation system while computing motion from first-principles physics rather than parametric tables.

(U) **Plugin File**: rotorcraft_6dof.dll (Windows) / librotorcraft_6dof.so (Linux)
AFSIM Type Name: Rotorcraft6DOF_Mover
Lines of Code: 2,939 across 6 source files

4.1.2 (U) Functional Behavior

(U) The 6DOF mover computes platform motion through the following physics pipeline executed each simulation timestep:

1. (U) **Navigation**: Read next waypoint from AFSIM route; compute desired position/velocity
2. (U) **Position Controller**: PD controller converts position error to desired velocity
3. (U) **Velocity Controller**: P controller converts velocity error to desired tilt angles
4. (U) **Attitude Controller**: PD controller converts attitude error to desired body moments
5. (U) **Rotor Model**: Compute individual rotor thrust and torque from throttle commands
6. (U) **Force Aggregation**: Sum all forces (gravity, rotor thrust, drag) in NED frame
7. (U) **Rigid Body Integration**: Semi-implicit Euler integration of Newton-Euler equations
8. (U) **State Update**: Update AFSIM platform position, velocity, and attitude

(U) **Rigid Body Equations of Motion**:

(U) Translational dynamics (Newton's second law in NED frame):

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$$\mathbf{F}_{net} = m\mathbf{a} = \mathbf{F}_{gravity} + R_{body \to NED} \cdot \mathbf{F}_{thrust} + \mathbf{F}_{drag} \quad (1)$$

(U) Translational Dynamics

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(U) Where:

- (U) m = total mass (kg)
- (U) $\mathbf{a}$ = acceleration in NED frame (m/s^2)
- (U) $R_{body \to NED}$ = rotation matrix from body to NED (derived from quaternion)
- (U) $\mathbf{F}_{gravity} = [0, 0, mg]^T$ (NED, down positive)

(U) Rotational dynamics (Euler's equation):

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$$\mathbf{M} = I\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (I\boldsymbol{\omega}) \quad (2)$$

(U) Rotational Dynamics

UNCLASSIFIED

(U) Where:

- (U) $\mathbf{M}$ = net moment vector (Nm)
- (U) I = inertia tensor (diagonal: I_{xx}, I_{yy}, I_{zz})
- (U) $\boldsymbol{\omega}$ = angular velocity vector (rad/s)

(U) **Attitude Representation**: Quaternion-based (avoids gimbal lock):

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$$\dot{q} = \frac{1}{2} q \otimes \boldsymbol{\omega}_q \tag{3}$$

(U) Quaternion Kinematic Equation

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(U) Where $\boldsymbol{\omega}_q = [0, \omega_x, \omega_y, \omega_z]$ is the angular velocity as a pure quaternion.

(U) **Integration Method**: Semi-implicit Euler (symplectic):

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$$\begin{aligned} \boldsymbol{\omega}(t + \Delta t) &= \boldsymbol{\omega}(t) + \dot{\boldsymbol{\omega}} \cdot \Delta t \\ q(t + \Delta t) &= q(t) + \dot{q}(t + \Delta t) \cdot \Delta t \\ \mathbf{v}(t + \Delta t) &= \mathbf{v}(t) + \mathbf{a} \cdot \Delta t \\ \mathbf{p}(t + \Delta t) &= \mathbf{p}(t) + \mathbf{v}(t + \Delta t) \cdot \Delta t \end{aligned} \tag{4}$$

(U) Semi-Implicit Euler Integration

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(U) This method preserves energy better than explicit Euler for oscillatory systems.

(U) **Rotor Thrust (Momentum Theory)**:

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$$T = C_T \cdot \rho \cdot n^2 \cdot D^4 \tag{5}$$

(U) Rotor Thrust

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(U) Where:

- (U) C_T = thrust coefficient (0.10)

- (U) ρ = air density (1.225 kg/m^3 at sea level)
- (U) n = rotational speed (rev/s = RPM/60)
- (U) D = propeller diameter (0.76 m (30 in))

(U) **Rotor Torque**:

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$$Q = C_Q \cdot \rho \cdot n^2 \cdot D^5 \tag{6}$$

(U) Rotor Torque

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(U) Where C_Q = torque coefficient (0.008).

4.1.3 (U) 6DOF Mover Key Performance Indicators

Table 6: (U) 6DOF Mover KPIs

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KPI	Measure	Expected	Validation
Position hold accuracy	Drift during hover (m)	< 1.0 m	Test 03: ~1e-17 m/s drift
Attitude recovery	Time to recover from disturbance	< 5 s	Test 02: Clean recovery
Velocity tracking	Steady-state error (m/s)	< 1.0 m/s	Test 04: 14.49 vs 15.0 m/s
Numerical stability	No NaN/divergence under stress	Bounded	Test 05: Stable under extremes
Max tilt angle	Maximum bank/pitch during maneuvers	25°	Test 05: 25° achieved

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4.1.4 (U) 6DOF Mover Parameters

Table 7: (U) 6DOF Mover Parameters

UNCLASSIFIED				
Parameter	Default	Units	Description	User Tunable
mass	123	kg	Total platform mass (MTOW)	Yes
inertia_xx	2.5	kg·m^2	Roll moment of inertia	Yes
inertia_yy	2.5	kg·m^2	Pitch moment of inertia	Yes
inertia_zz	4.5	kg·m^2	Yaw moment of inertia	Yes
rotor_diameter	0.76	m	Propeller diameter	Yes
arm_length	0.5	m	Motor arm length from CG	Yes
thrust_coefficient	0.10	–	C_T for momentum theory	Yes
torque_coefficient	0.008	–	C_Q for reaction torque	Yes
max_rpm	4500	RPM	Maximum motor RPM	Yes
max_speed	30	m/s	Maximum horizontal speed	Yes
max_climb_rate	8	m/s	Maximum climb rate	Yes
max_descent_rate	4	m/s	Maximum descent rate	Yes
max_tilt_angle	25	deg	Maximum bank/pitch angle	Yes
altitude_kp	0.3	–	Altitude PID proportional gain	Yes
altitude_ki	0.05	–	Altitude PID integral gain	Yes
altitude_kd	1.2	–	Altitude PID derivative gain	Yes
position_kp	1.0	–	Position PD proportional gain	Yes
position_kd	0.1	–	Position PD derivative gain	Yes
velocity_kp	3.0	–	Velocity P gain	Yes
attitude_kp	8.0	–	Attitude PD proportional gain	Yes
attitude_kd	2.0	–	Attitude PD derivative gain	Yes
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4.1.5 (U) 6DOF Mover Assumptions, Limitations, and Gaps

(U) **Assumptions**:

- (U) Rigid body dynamics (no structural flexibility)
- (U) Constant air density (no altitude-dependent atmospheric model)
- (U) Symmetric inertia tensor (diagonal only; no cross-coupling products)
- (U) Instantaneous force application (no actuator dynamics beyond motor time constant)

(U) **Limitations**:

- (U) Maximum tilt angle capped at 25 degrees (actual aircraft may achieve more aggressive attitudes)
- (U) No wind model — all flight occurs in still air
- (U) No sensor noise injection (perfect state knowledge for controllers)
- (U) Coaxial rotor interference modeled as a constant factor (0.85), not dynamic

(U) **Gaps**:

- (U) Inertia values are estimated, not measured from actual T-150 hardware
- (U) No BEMT (Blade Element Momentum Theory) rotor model — uses simplified momentum theory
- (U) No structural vibration or aeroelastic effects

4.2 (U) Battery Processor

4.2.1 (U) Component Overview

(U) The T150_BATTERY_PROCESSOR models a 13S Lithium-Ion battery pack with state-of-charge tracking, open-circuit voltage modeling, voltage sag under load, and SOC-dependent internal resistance. It reads dynamic power consumption from the 6DOF mover or Propulsion Processor via AuxData and computes real-time battery state.

(U) **Plugin File**: t150_battery.dll (Windows) / libt150_battery_processor.so (Linux)
AFSIM Type Name: T150_BATTERY_PROCESSOR
Lines of Code: 1,139

4.2.2 (U) Functional Behavior

(U) The Battery Processor executes the following computation each update cycle:

1. (U) **Read Power Demand**: Read 6dof_total_power or PROP_TOTAL_POWER from AuxData (falls back to configured hover_power_w if unavailable)
2. (U) **Apply Efficiency**: Divide by motor efficiency to get electrical power
3. (U) **Add Auxiliary Load**: Add avionics/cooling power
4. (U) **Compute OCV**: Calculate open-circuit voltage from SOC using polynomial fit
5. (U) **Compute Internal Resistance**: Apply SOC-dependent resistance multiplier
6. (U) **Solve for Current**: Use quadratic formula to find stable current solution
7. (U) **Compute Terminal Voltage**: OCV minus IR drop
8. (U) **Update Energy Consumed**: Integrate power over timestep
9. (U) **Update SOC**: Calculate remaining state of charge
10. (U) **Check Warnings**: Evaluate low-battery and voltage sag thresholds
11. (U) **Write to AuxData**: Publish all battery state variables

(U) **Open-Circuit Voltage (OCV) Model**:

(U) The OCV follows a polynomial fit to a typical Li-Ion discharge curve:

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$$V_{cell} = a_0 + a_1 \cdot SOC + a_2 \cdot SOC^2 + a_3 \cdot SOC^3 \tag{7}$$

(U) Cell Open-Circuit Voltage

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(U) Where:

- (U) $a_0 = 3.0$ V (cutoff voltage per cell)
- (U) $a_1 = 1.2$ (linear term)
- (U) $a_2 = -0.5$ (creates flat middle region)
- (U) $a_3 = 0.5$ (steeper at ends)

(U) Pack voltage:

UNCLASSIFIED

$$V_{pack} = V_{cell} \times 13 \tag{8}$$

(U) Pack Open-Circuit Voltage

UNCLASSIFIED

(U) **Voltage Sag Model**:

(U) Under load, terminal voltage drops due to internal resistance:

UNCLASSIFIED

$$V_{terminal} = V_{OCV} - I \cdot R_{internal} \tag{9}$$

(U) Terminal Voltage Under Load

UNCLASSIFIED

(U) Since $P = V_{terminal} \cdot I$, combining equations yields a quadratic:

UNCLASSIFIED

$$I = \frac{V_{OCV} - \sqrt{V_{OCV}^2 - 4 \cdot R \cdot P}}{2 \cdot R} \tag{10}$$

(U) Voltage Sag Quadratic Solution

UNCLASSIFIED

(U) The smaller root is selected as the physically meaningful solution (lower current, higher efficiency).

(U) **SOC-Dependent Internal Resistance**:

(U) Internal resistance follows a U-shaped curve:

UNCLASSIFIED

$$R_{eff} = R_{base} \times f(SOC) \tag{11}$$

(U) Effective Internal Resistance

UNCLASSIFIED

(U) Where:

- (U) SOC > 0.6: $f = 1.0 + 0.3 \times \frac{(SOC - 0.6)^2}{0.16}$ (peaks ~1.3x at 100%)
- (U) $0.2 <$ SOC < 0.6: $f = 1.0$ (minimum in middle range)
- (U) SOC < 0.2: $f = 1.0 + 2.0 \times \frac{(0.2 - SOC)^2}{0.04}$ (peaks ~3.0x at 0%)

(U) **Energy Consumption**:

UNCLASSIFIED

$$E_{consumed}(t) = E_{consumed}(t-1) + P_{total} \cdot \frac{\Delta t}{3600}$$
$$SOC = 1.0 - \frac{E_{consumed}}{E_{capacity}} \tag{12}$$

(U) Energy Consumption and SOC Update

UNCLASSIFIED

4.2.3 (U) Battery Processor Key Performance Indicators

Table 8: (U) Battery Processor KPIs

UNCLASSIFIED

KPI	Measure	Expected	Validation
Energy conservation	SOC tracking error	$< 0.01\%$	Test 09: $< 0.01\%$
Voltage sag at full load	Voltage drop (V)	5–15 V	Test 06: Validated
Low SOC tracking	Correct SOC from non-100% start	Monotonic decrease	Test 06: Fixed and validated
Flight endurance	Time to battery depletion	14–36 min	Test 10: 14.3 min
Warning triggers	Low battery / voltage warnings	At configured thresholds	Test 10: All triggered

UNCLASSIFIED

4.2.4 (U) Battery Processor Parameters

Table 9: (U) Battery Processor Parameters

UNCLASSIFIED

Parameter	Default	Units	Description	User Tunable
battery_capacity_wh	7000.0	Wh	Total battery energy capacity	Yes

UNCLASSIFIED

Continued on next page

Table 9: (U) Battery Processor Parameters (Continued)

UNCLASSIFIED				
Parameter	Default	Units	Description	User Tunable
initial_soc	1.0	0–1	Initial state of charge	Yes
hover_power_w	18000.0	W	Fallback hover power (if dynamic unavailable)	Yes
motor_efficiency	0.85	0–1	Motor/ESC electrical efficiency	Yes
auxiliary_power_w	4000.0	W	Avionics, cooling, other loads	Yes
low_battery_warning	0.20	0–1	SOC threshold for low-battery warning	Yes
critical_battery_level	0.10	0–1	SOC threshold for critical warning	Yes
nominal_voltage	46.8	V	Pack nominal voltage (13S × 3.6V)	Yes
full_charge_voltage	54.6	V	Pack full charge voltage (13S × 4.2V)	Yes
cutoff_voltage	39.0	V	Pack cutoff voltage (13S × 3.0V)	Yes
internal_resistance	0.025	Ω	Pack internal resistance	Yes
voltage_sag_warning	42.0	V	Voltage threshold for sag warning	Yes
use_dynamic_power	true	bool	Read power from mover/propulsion AuxData	Yes
power_source_key	6dof_total_power	string	AuxData key for dynamic power input	Yes
csv_output_file	–	path	Optional CSV logging file path	Yes
update_interval	0.1	sec	Processor update rate	Yes
UNCLASSIFIED				

4.2.5 (U) Battery Processor Assumptions, Limitations, and Gaps

(U) **Assumptions**:

- (U) Li-Ion chemistry with characteristic polynomial discharge curve
- (U) Uniform cell aging across all 13 series cells
- (U) Constant temperature effects on internal resistance (no thermal feedback from environment)
- (U) Motor efficiency is constant across operating range

(U) **Limitations**:

- (U) No battery degradation over charge/discharge cycles
- (U) No temperature-dependent capacity derating
- (U) Internal resistance model is simplified (no Randles circuit or EIS-based model)
- (U) No cell-level balancing effects modeled

(U) **Gaps**:

- (U) Actual T-150 battery pack resistance not measured; 25 mΩ is estimated for pack-level resistance
- (U) OCV polynomial coefficients are generic Li-Ion, not specific to T-150 cell chemistry
- (U) Auxiliary power (4 kW) may be high; typical avionics draw 0.5–2 kW

4.3 (U) Propulsion Processor

4.3.1 (U) Component Overview

(U) The T150_PROPULSION_PROCESSOR models the 8-motor coaxial propulsion system including individual motor state tracking (RPM, temperature, efficiency), thrust calculation via momentum theory, motor thermal dynamics, ground effect thrust augmentation, and payload mass management.

(U) **Plugin File**: t150_propulsion.dll (Windows) / libt150_propulsion.so (Linux)
AFSIM Type Name: T150_PROPULSION_PROCESSOR
Lines of Code: 1,281

4.3.2 (U) Functional Behavior

(U) The Propulsion Processor computes the following each update cycle:

1. (U) **Motor RPM**: Calculate target RPM from voltage and KV rating; apply first-order lag
2. (U) **Motor Thrust**: Compute per-motor thrust using momentum theory
3. (U) **Motor Torque**: Compute per-motor reaction torque
4. (U) **Motor Efficiency**: Parabolic model based on RPM ratio
5. (U) **Electrical Power**: Mechanical power divided by efficiency
6. (U) **Thermal Update**: Compute heat generation and dissipation
7. (U) **Ground Effect**: Apply Cheeseman-Bennett thrust augmentation if near ground
8. (U) **Payload**: Track total mass and CG effects
9. (U) **Dynamic Power**: Calculate total power from flight state (velocity, climb, bank)
10. (U) **Write AuxData**: Publish all propulsion state variables

(U) **Motor RPM with First-Order Lag**:

UNCLASSIFIED

$$\begin{aligned} RPM_{target} &= KV \times V_{nominal} \times 0.85 \\ \alpha &= \frac{\Delta t}{\tau + \Delta t} \\ RPM_{new} &= RPM_{old} + \alpha \times (RPM_{target} - RPM_{old}) \end{aligned} \tag{13}$$

(U) Motor RPM First-Order Lag

UNCLASSIFIED

(U) Where $\tau = 0.15s$ (150 ms time constant for U15II with 30" propeller).

(U) **Motor Efficiency (Parabolic Model)**:

UNCLASSIFIED

$$\begin{aligned} \eta_{RPM} &= 1.0 - 0.5 \times (RPM_{ratio} - 0.7)^2 \\ \eta_{torque} &= 1.0 - 0.2 \times Torque_{ratio} \\ \eta &= \eta_{peak} \times \eta_{RPM} \times \eta_{torque} \end{aligned} \tag{14}$$

(U) Motor Efficiency Model

UNCLASSIFIED

(U) Peak efficiency occurs at approximately 70% of maximum RPM.

(U) **Thermal Model**:

UNCLASSIFIED

$$\begin{aligned} \dot{Q}_{in} &= P_{electrical} \times (1 - \eta) \\ \dot{Q}_{out} &= k_{cooling} \times (T_{motor} - T_{ambient}) \\ \frac{dT}{dt} &= \frac{\dot{Q}_{in} - \dot{Q}_{out}}{C_{thermal}} \end{aligned} \tag{15}$$

(U) Motor Thermal Model

UNCLASSIFIED

(U) Where:

- (U) $C_{thermal} = 1500J/K$ (motor thermal mass)
- (U) $k_{cooling} = 20W/K$ (forced convection from prop wash)

(U) **Ground Effect (Cheeseman-Bennett Approximation)**:

UNCLASSIFIED

$$\frac{T_{IGE}}{T_{OGE}} = \frac{1}{1 - \left(\frac{R}{4h}\right)^2} \tag{16}$$

(U) Ground Effect – Cheeseman-Bennett

UNCLASSIFIED

(U) Where:

- (U) R = rotor radius (0.38 m)
- (U) h = rotor height above ground

(U) Active below 1 rotor diameter (0.76 m) AGL. Clamped to maximum 30% thrust boost.

(U) **Dynamic Power Model (Phase 6)**:

UNCLASSIFIED

$$P_{total} = P_{induced} + P_{climb} + P_{parasite} + P_{accel} \tag{17}$$

(U) Total Dynamic Power

UNCLASSIFIED

Table 10: (U) Dynamic Power Components

UNCLASSIFIED			
Component	Symbol	Equation	Notes
Induced Power	$P_{induced}$	$k \cdot T \cdot v_i$ where $v_i = \sqrt{T/(2\rho A)}$	Momentum theory
Climb Power	P_{climb}	$m \cdot g \cdot V_{climb}$	Weight × vertical speed
Parasite Power	$P_{parasite}$	$0.5 \cdot \rho \cdot C_d \cdot A_f \cdot V^3$	Drag at speed
Acceleration Power	P_{accel}	$m \cdot a \cdot V$	Transient maneuvers
			UNCLASSIFIED

(U) Thrust adjustment for bank angle (coordinated turns):

UNCLASSIFIED

$$T_{required} = \frac{m \cdot g}{\cos(\phi)}$$
$$\phi = \arctan\left(\frac{a_{horizontal}}{g}\right) \tag{18}$$

(U) Bank Angle Thrust Adjustment

UNCLASSIFIED

(U) Where ϕ = bank angle estimated from horizontal acceleration.

4.3.3 (U) Propulsion Processor Key Performance Indicators

Table 11: (U) Propulsion Processor KPIs

UNCLASSIFIED			
KPI	Measure	Expected	Validation
Hover power	Electrical power in hover	17–25 kW	Test 08: 19.5–20.1 kW mechanical
Motor temperature	Equilibrium temperature	60–90°C	Test 01: Stable
Ground effect factor	Thrust multiplier near ground	1.0–1.3	Test 07: Functional
Power vs. speed	Power variation with airspeed	Cruise < hover	Validated (pattern tests)
			UNCLASSIFIED

4.3.4 (U) Propulsion Processor Parameters

Table 12: (U) Propulsion Processor Parameters

UNCLASSIFIED				
Parameter	Default	Units	Description	User Tunable
motor_count	8	–	Number of motors	Yes
kv_rating	100.0	RPM/V	Motor KV rating	Yes
				UNCLASSIFIED

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Table 12: (U) Propulsion Processor Parameters (Continued)

UNCLASSIFIED				
Parameter	Default	Units	Description	User Tunable
max_power_per_ motor	3500.0	W	Maximum power per motor	Yes
propeller_diameter	0.76	m	Propeller diameter (30 in)	Yes
coaxial_efficiency	0.85	0–1	Coaxial rotor interference factor	Yes
motor_thermal_limit	120.0	°C	Motor overtemperature threshold	Yes
motor_thermal_mass	1500.0	J/K	Motor thermal mass	Yes
motor_cooling_rate	20.0	W/K	Forced convection cooling rate	Yes
motor_time_constant	0.15	s	RPM response time constant	Yes
ground_effect_ enabled	true	bool	Enable ground effect modeling	Yes
empty_mass	68.0	kg	Empty aircraft mass	Yes
payload_mass	0.0	kg	Current payload mass	Yes
max_payload_mass	55.0	kg	Maximum payload capacity	Yes
update_interval	0.1	sec	Processor update rate	Yes
UNCLASSIFIED				

4.3.5 (U) Propulsion Processor Assumptions, Limitations, and Gaps

(U) **Assumptions**:

- (U) All 8 motors operate identically (no individual motor failures modeled)
- (U) Coaxial interference is a constant factor (0.85), not RPM/inflow-dependent
- (U) Air density is constant (sea-level ISA, 1.225 kg/m^3)
- (U) Motor efficiency peaks at 70% RPM (generic assumption for large brushless motors)

(U) **Limitations**:

- (U) No individual motor failure or degradation modes
- (U) Propeller aerodynamics use simplified momentum theory (no BEMT)
- (U) No blade pitch variation (fixed-pitch propellers)
- (U) Ground effect uses single-rotor approximation (does not model multi-rotor ground effect interaction)

(U) **Gaps**:

- (U) Motor thermal parameters estimated from motor class, not specific U15II test data
- (U) Coaxial interference factor (0.85) is generic; actual T-150 interference may differ
- (U) No wind/gust model affects propulsion performance

4.4 (U) VRS Processor

4.4.1 (U) Component Overview

(U) The T150_VRS_PROCESSOR detects and models Vortex Ring State (VRS) conditions, a dangerous aerodynamic phenomenon where a rotorcraft descends into its own rotor downwash, causing severe loss of lift and control authority. The processor monitors descent rate and forward speed, implements a 5-state machine, and computes lift degradation and control authority factors.

(U) **Plugin File**: t150_vrs.dll (Windows) / libt150_vrs.so (Linux)
AFSIM Type Name: T150_VRS_PROCESSOR
Lines of Code: 697

4.4.2 (U) Functional Behavior

(U) The VRS Processor reads platform velocity from the AFSIM mover each update cycle and evaluates VRS risk:

1. (U) **Read Velocity**: Get NED velocity from platform mover
2. (U) **Compute Descent Rate**: $V_d = \max(0, V_{down})$ (positive = descending)
3. (U) **Compute Forward Speed**: $V_{fwd} = \sqrt{V_N^2 + V_E^2}$
4. (U) **Compute VRS Margin**: Combined margin from descent rate and forward speed
5. (U) **Update State Machine**: Transition between 5 VRS states
6. (U) **Compute Degradation**: Calculate lift and control authority factors
7. (U) **Write AuxData**: Publish VRS state, margin, and factors

(U) **VRS Margin Calculation**:

(U) Descent margin:

UNCLASSIFIED

$$m_{descent} = \begin{cases} 1.0 & \text{if } V_d \leq 0 \\ 1.0 - V_d/V_{threshold} & \text{if } 0 < V_d < V_{threshold} \\ -(V_d - V_{threshold})/V_{threshold} & \text{if } V_d \geq V_{threshold} \end{cases} \quad (19)$$

(U) VRS Descent Margin

UNCLASSIFIED

(U) Speed margin:

UNCLASSIFIED

$$m_{speed} = \min(1.0,\ V_{fwd}/V_{speed_threshold}) \tag{20}$$

(U) VRS Speed Margin

UNCLASSIFIED

(U) Combined margin:

UNCLASSIFIED

$$m = \begin{cases} 1.0 & \text{if } V_{fwd} \geq V_{speed_threshold} \\ m_{descent} \cdot w + m_{speed} \cdot (1 - w) & \text{otherwise} \end{cases} \tag{21}$$

(U) VRS Combined Margin

UNCLASSIFIED

(U) Where $w = 1.0 - m_{speed}$ (weighting factor).

(U) **State Machine**:

(U) The VRS processor implements a 5-state machine governing transitions between flight safety states:

Table 13: (U) VRS State Machine Transitions

UNCLASSIFIED		
From State	To State	Condition
CLEAR	CAUTION	margin < 0.8
CAUTION	CLEAR	margin ≥ 1.0
CAUTION	ONSET	margin < 0.5
ONSET	RECOVERY	margin ≥ 0.8
ONSET	DEVELOPED	margin < 0.2 for 2 seconds
DEVELOPED	RECOVERY	margin ≥ 0.5
		UNCLASSIFIED

Continued on next page

Table 13: (U) VRS State Machine Transitions (Continued)

UNCLASSIFIED		
From State	To State	Condition
RECOVERY	CLEAR	development cleared
		UNCLASSIFIED

(U) **Performance Degradation Factors**:

Table 14: (U) VRS Performance Degradation Factors

UNCLASSIFIED		
State	Lift Factor	Control Factor
CLEAR	1.00	1.00
CAUTION	0.95	1.00
ONSET	0.85	0.90
DEVELOPED	0.40	0.50
RECOVERY	Proportional to development	Proportional
		UNCLASSIFIED

4.4.3 (U) VRS Processor Parameters

Table 15: (U) VRS Processor Parameters

UNCLASSIFIED				
Parameter	Default	Units	Description	User Tunable
vrs_descent_ threshold	4.0	m/s	Descent rate for VRS risk	Yes
vrs_speed_threshold	5.0	m/s	Forward speed for VRS immunity	Yes
caution_margin	0.8	0–1	Margin threshold for CAUTION state	Yes
induced_velocity	8.5	m/s	Rotor induced velocity in hover	Yes
coaxial_factor	0.85	0–1	Coaxial VRS resistance factor	Yes
update_interval	0.05	sec	High-rate update for VRS detection (20 Hz)	Yes
				UNCLASSIFIED

4.4.4 (U) VRS Processor Assumptions, Limitations, and Gaps

(U) **Assumptions**:

- (U) VRS boundaries based on momentum theory induced velocity ratio
- (U) Coaxial configuration provides inherent VRS resistance (factor 0.85)
- (U) Recovery requires either forward speed increase or descent rate reduction

(U) **Limitations**:

- (U) VRS is detected but does not feed back to modify mover dynamics (informational only)
- (U) No turbulence-induced VRS modeling
- (U) Development time (2s to full VRS) is estimated, not measured

(U) **Gaps**:

- (U) No coaxial-specific VRS boundary data; factor is estimated
- (U) Lift and control degradation factors are representative, not from T-150 flight test

4.5 (U) Slung Load Processor

4.5.1 (U) Component Overview

(U) The T150_SLUNG_LOAD_PROCESSOR models cable-suspended payload dynamics using a 2-DOF spherical pendulum model. It computes swing angles, cable tension, and reaction forces on the aircraft from suspended load motion excited by aircraft maneuvers.

(U) **Plugin File**: t150_slung_load.dll (Windows) / libt150_slung_load.so (Linux)
AFSIM Type Name: T150_SLUNG_LOAD_PROCESSOR
Lines of Code: 825

4.5.2 (U) Functional Behavior

(U) The Slung Load Processor integrates pendulum dynamics at 50 Hz:

1. (U) **Read Aircraft State**: Get velocity and compute acceleration from mover
2. (U) **Compute Excitation**: Aircraft acceleration drives pendulum
3. (U) **Integrate Pendulum Equations**: Semi-implicit Euler at 50 Hz
4. (U) **Compute Cable Tension**: Static + centripetal components
5. (U) **Compute Reaction Forces**: Forces on aircraft from load
6. (U) **Check States**: Detect resonance, overload, or release conditions
7. (U) **Write AuxData**: Publish all slung load state variables

(U) **Spherical Pendulum Equations of Motion**:

UNCLASSIFIED

$$\ddot{\theta}_x = -\frac{g}{L}\sin(\theta_x) - \frac{a_x}{L}\cos(\theta_x) - c \cdot \dot{\theta}_x$$
$$\ddot{\theta}_y = -\frac{g}{L}\sin(\theta_y) - \frac{a_y}{L}\cos(\theta_y) - c \cdot \dot{\theta}_y \quad (22)$$

(U) Spherical Pendulum Equations of Motion

UNCLASSIFIED

(U) Where:

- (U) θ_x, θ_y = swing angles in X and Y planes (rad)

- (U) $g = 9.81$ m/s^2
- (U) $L =$ cable length (m)
- (U) a_x, $a_y =$ aircraft acceleration (excitation source)
- (U) $c =$ cable damping coefficient (1/s)

(U) **Natural Frequency**:

UNCLASSIFIED

$$f_n = \frac{1}{2\pi}\sqrt{\frac{g}{L}} \tag{23}$$

(U) Pendulum Natural Frequency

UNCLASSIFIED

(U) For a 5 m cable: $f_n = 0.223$ Hz (4.5 second period).

(U) **Cable Tension**:

UNCLASSIFIED

$$T = m \cdot (g\cos(\theta) - a_z + L \cdot \omega^2) \tag{24}$$

(U) Cable Tension

UNCLASSIFIED

(U) Where $\omega = \sqrt{\dot{\theta}_x^2 + \dot{\theta}_y^2}$ (angular velocity magnitude).

(U) Constraints: $T \geq 0$ (cable cannot push) and $T \leq T_{max}$ (cable strength limit).

(U) **Reaction Forces on Aircraft** (body frame):

UNCLASSIFIED

$$F_x = T\sin(\theta_x), \quad F_y = T\sin(\theta_y), \quad F_z = T\cos(\theta) \tag{25}$$

(U) Slung Load Reaction Forces

UNCLASSIFIED

4.5.3 (U) Slung Load Processor Parameters

Table 16: (U) Slung Load Processor Parameters

UNCLASSIFIED

Parameter	Default	Units	Description	User Tunable
load_mass	0.0	kg	Suspended load mass	Yes
cable_length	5.0	m	Cable length	Yes
cable_damping	0.1	1/s	Air/friction damping coefficient	Yes
attach_point_z	0.5	m	Attach point below CG	Yes
max_cable_tension	5000.0	N	Maximum cable tension before failure	Yes
resonance_threshold	20.0	deg	Swing angle for resonance warning	Yes
update_interval	0.02	sec	Dynamics update rate (50 Hz)	Yes

UNCLASSIFIED

4.5.4 (U) Slung Load Processor Assumptions, Limitations, and Gaps

(U) **Assumptions**:

- (U) Cable is massless and inextensible
- (U) Load is a point mass (no aerodynamic drag on load)
- (U) Small-angle behavior for linearized natural frequency
- (U) Cable damping is constant (not velocity-dependent)

(U) **Limitations**:

- (U) 2-DOF spherical pendulum only (no cable elasticity or bending)
- (U) No aerodynamic drag on the suspended load
- (U) No load release/pickup dynamics
- (U) Reaction forces are informational; do not feed back to mover dynamics

(U) **Gaps**:

- (U) No T-150-specific cable data; damping coefficient is estimated
- (U) Cable breaking strength (5000 N) is generic
- (U) No multi-point suspension modeling

5 (U) Model Setup and Configuration

5.1 (U) Scenario Configuration

(U) The T-150 model is configured through AFSIM scenario files (.afsim). There is no GUI; all parameters are set in text-based scenario scripts.

(U) **Note**: This section replaces the ODESSA GUI setup section. AFSIM uses text-based scenario configuration rather than a graphical interface.

5.1.1 (U) Installation

```
# Clone the repository
git clone https://github.com/DSGarrett/project005_t150_uas.git
cd project005_t150_uas

# Build plugins
./ngic_afsim_build.sh

# Install to AFSIM (requires elevated privileges)
sudo ./ngic_afsim_install.sh
```

(U) Plugins are installed to:

- (U) Linux: /opt/afsim*/lib/wsf_plugins/ and /opt/afsim*/bin/mission_plugins/
- (U) Windows: C:\afsim\AFSIM-2.2505\lib\wsf_plugins\ and C:\afsim\AFSIM-2.2505\bin\mission_plugins\

5.1.2 (U) Minimal Platform Definition

```
platform_type T150_BASIC WSF_PLATFORM
    icon t150

    mover Rotorcraft6DOF_Mover
        mass 123 kg
        max_speed 30 m/s
    end_mover

    processor BATTERY T150_BATTERY_PROCESSOR
        battery_capacity_wh 7000
        use_dynamic_power true
    end_processor

end_platform_type
```

5.1.3 (U) Full Configuration Example

```
platform_type T150_FULL_PHYSICS T150_UAS

    // Battery with voltage sag model
    processor BATTERY_MONITOR T150_BATTERY_PROCESSOR
        battery_capacity_wh     7000.0
        initial_soc             1.0
        nominal_voltage         46.8
        internal_resistance     0.025
        use_dynamic_power       true
    end_processor

    // Propulsion with thermal and ground effect
    processor PROPULSION_MONITOR T150_PROPULSION_PROCESSOR
        motor_count             8
        kv_rating               100
        max_power_per_motor     3500
        propeller_diameter      0.76
        coaxial_efficiency      0.85
        motor_thermal_limit     120
        ground_effect_enabled   true
        empty_mass              68.0
        payload_mass            40.0
    end_processor

    // VRS detection
    processor VRS_MONITOR T150_VRS_PROCESSOR
        vrs_descent_threshold   4.0
        vrs_speed_threshold     5.0
        coaxial_factor          0.85
    end_processor

    // Slung load dynamics
    processor SLUNG_LOAD T150_SLUNG_LOAD_PROCESSOR
        load_mass               25.0
        cable_length            6.0
        cable_damping           0.1
    end_processor

end_platform_type
```

5.2 (U) Model Parameters (Master Table)

(U) The following table provides the complete list of all configurable parameters across all plugins:

Table 17: (U) Master Model Parameters

UNCLASSIFIED					
Variable	Description	Default	Component	User Tunable?	Source
mass	Total platform mass	(U) 123 kg	6DOF Mover	Yes	[4]
					UNCLASSIFIED

Continued on next page

Table 17: (U) Master Model Parameters (Continued)

UNCLASSIFIED

Variable	Description	Default	Component	User Tunable?	Source
inertia_xx	Roll moment of inertia	(U) 2.5 kg·m^2	6DOF Mover	Yes	Est.
inertia_yy	Pitch moment of inertia	(U) 2.5 kg·m^2	6DOF Mover	Yes	Est.
inertia_zz	Yaw moment of inertia	(U) 4.5 kg·m^2	6DOF Mover	Yes	Est.
rotor_diameter	Propeller diameter	(U) 0.76 m	6DOF Mover	Yes	[6]
thrust_ coefficient	C_T for momentum theory	(U) 0.10	6DOF Mover	Yes	[9]
max_speed	Max horizontal speed	(U) 30 m/s	6DOF Mover	Yes	[11]
battery_ capacity_wh	Total battery capacity	(U) 7000 Wh	Battery	Yes	[5]
nominal_ voltage	Pack nominal voltage	(U) 46.8 V	Battery	Yes	[5]
internal_ resistance	Pack resistance	(U) 0.025 Ω	Battery	Yes	Est.
motor_ efficiency	Motor/ESC efficiency	(U) 0.85	Battery	Yes	[6]
motor_count	Number of motors	(U) 8	Propulsion	No	[4]
kv_rating	Motor KV rating	(U) 100 RPM/V	Propulsion	Yes	[6]
coaxial_ efficiency	Coaxial interference	(U) 0.85	Propulsion	Yes	[9]
motor_ thermal_mass	Motor thermal mass	(U) 1500 J/K	Propulsion	Yes	Est.
motor_ cooling_rate	Cooling rate	(U) 20 W/K	Propulsion	Yes	Est.
vrs_descent_ threshold	VRS descent rate	(U) 4.0 m/s	VRS	Yes	[10]
vrs_speed_ threshold	VRS forward speed	(U) 5.0 m/s	VRS	Yes	[10]
load_mass	Slung load mass	(U) 0.0 kg	Slung Load	Yes	Scenario
cable_length	Cable length	(U) 5.0 m	Slung Load	Yes	Scenario

UNCLASSIFIED

Continued on next page

Table 17: (U) Master Model Parameters (Continued)

UNCLASSIFIED					
Variable	Description	Default	Component	User Tunable?	Source
cable_damping	Cable damping	(U) 0.1 1/s	Slung Load	Yes	Est.
					UNCLASSIFIED

5.3 (U) Model Specific Log Files

(U) The simulation can generate the following output files:

Table 18: (U) Model Log Files

UNCLASSIFIED			
Output	Format	Content	Configuration
Battery CSV	CSV	SOC, voltage, current, power per timestep	csv_output_file parameter
AFSIM AER	AER (binary)	Full simulation event records	event_pipe in scenario
AFSIM Log	Text	Console output with processor messages	log_file in scenario
Telemetry CSV	CSV	6DOF state: position, velocity, attitude	Optional 6DOF logging
			UNCLASSIFIED

6 (U) Analytic Outputs

6.1 (U) AuxData Output Keys

(U) All model state is published to the AFSIM AuxData container, accessible from scenario scripts and other processors. The following tables list all output keys by component.

(U) **Battery Processor Outputs**:

Table 19: (U) Battery Processor AuxData Outputs

UNCLASSIFIED			
Key	Type	Units	Description
BATTERY_SOC	double	0–1	State of charge
BATTERY_POWER_W	double	W	Current power draw
BATTERY_ENERGY_WH	double	Wh	Remaining energy
BATTERY_VOLTAGE	double	V	Terminal voltage under load
BATTERY_OCV	double	V	Open-circuit voltage
BATTERY_VOLTAGE_SAG	double	V	Voltage drop from IR
BATTERY_CURRENT_A	double	A	Current draw
BATTERY_LOW	bool	–	Low battery warning flag
BATTERY_VOLTAGE_LOW	bool	–	Voltage sag warning flag
			UNCLASSIFIED

(U) **Propulsion Processor Outputs**:

Table 20: (U) Propulsion Processor AuxData Outputs

UNCLASSIFIED			
Key	Type	Units	Description
PROP_TOTAL_THRUST	double	N	Total thrust output
			UNCLASSIFIED

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Table 20: (U) Propulsion Processor AuxData Outputs (Continued)

UNCLASSIFIED			
Key	Type	Units	Description
PROP_TOTAL_POWER	double	W	Total electrical power
PROP_SYSTEM_EFF	double	0–1	System efficiency
PROP_MOTOR_COUNT	int	–	Operational motor count
PROP_MAX_TEMP	double	°C	Hottest motor temperature
PROP_AVG_TEMP	double	°C	Average motor temperature
PROP_GROUND_EFFECT	double	1.0–1.3	Ground effect factor
PROP_ALTITUDE_AGL	double	m	Altitude above ground
PROP_TOTAL_MASS	double	kg	Aircraft + payload mass
PROP_PAYLOAD_MASS	double	kg	Payload mass
PROP_THRUST_MARGIN	double	0–1	Available thrust reserve
PROP_CG_OFFSET_X	double	m	CG offset X
PROP_CG_OFFSET_Y	double	m	CG offset Y
PROP_CG_OFFSET_Z	double	m	CG offset Z
PROP_OVERLOADED	bool	–	Payload exceeds capacity
PROP_FORWARD_SPEED	double	m/s	Horizontal speed
PROP_CLIMB_RATE	double	m/s	Vertical speed (positive=up)
PROP_BANK_ANGLE	double	deg	Estimated bank angle
PROP_ACCEL_MAG	double	m/s^2	Horizontal acceleration
			UNCLASSIFIED

(U) **VRS Processor Outputs**:

Table 21: (U) VRS Processor AuxData Outputs

UNCLASSIFIED			
Key	Type	Units	Description
VRS_MARGIN	double	0–1	Safety margin (0=VRS, 1=safe)
VRS_STATE	int	0–4	State (CLEAR=0, CAUTION=1, ONSET=2, DEVELOPED=3, RECOVERY=4)
VRS_LIFT_FACTOR	double	0.4–1.0	Lift degradation factor
VRS_CONTROL_FACTOR	double	0.5–1.0	Control authority factor
VRS_DEVELOPMENT	double	0–1	VRS development level
VRS_DESCENT_RATE	double	m/s	Current descent rate
VRS_FORWARD_SPEED	double	m/s	Current forward speed
VRS_CAUTION	bool	–	Caution warning flag
VRS_ONSET	bool	–	Onset warning flag
VRS_CRITICAL	bool	–	Critical (developed VRS) flag
			UNCLASSIFIED

(U) **Slung Load Processor Outputs**:

Table 22: (U) Slung Load Processor AuxData Outputs

UNCLASSIFIED			
Key	Type	Units	Description
SLUNG_LOAD_STATE	int	0–4	Load state (DETACHED=0, ATTACHED=1, SWINGING=2, RESONANCE=3, RELEASED=4)
SLUNG_LOAD_MASS	double	kg	Load mass
SLUNG_CABLE_LENGTH	double	m	Cable length
SLUNG_CABLE_TENSION	double	N	Cable tension
SLUNG_SWING_X	double	deg	Swing angle X plane
SLUNG_SWING_Y	double	deg	Swing angle Y plane
SLUNG_SWING_MAG	double	deg	Total swing magnitude
SLUNG_FORCE_X	double	N	Reaction force on aircraft X
SLUNG_FORCE_Y	double	N	Reaction force on aircraft Y
			UNCLASSIFIED

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Table 22: (U) Slung Load Processor AuxData Outputs (Continued)

UNCLASSIFIED			
Key	Type	Units	Description
SLUNG_FORCE_Z	double	N	Reaction force on aircraft Z
SLUNG_POS_X	double	m	Load position X relative to aircraft
SLUNG_POS_Y	double	m	Load position Y
SLUNG_POS_Z	double	m	Load position Z
SLUNG_NATURAL_FREQ	double	Hz	Natural frequency
SLUNG_RESONANCE	bool	–	Resonance warning flag
			UNCLASSIFIED

6.2 (U) Analytic Use of Outputs

(U) **Flight Endurance Analysis**: Monitor BATTERY_SOC over time to determine total flight time under specific mission profiles. Compare energy consumption rates across different flight patterns.

(U) **Vulnerability Assessment**: Monitor VRS_MARGIN and VRS_STATE during descent maneuvers to identify vulnerable flight regimes. Low margins indicate flight conditions where the platform is susceptible to loss of control.

(U) **Payload Impact Analysis**: Compare PROP_THRUST_MARGIN and BATTERY_SOC depletion rates with and without payload to quantify the operational cost of payload carriage.

(U) **Slung Load Stability**: Monitor SLUNG_SWING_MAG and SLUNG_CABLE_TENSION during maneuvers to identify flight profiles that induce dangerous load oscillations.

7 (U) Sources, Assumptions, Limitations, and Gaps

7.1 (U) Sources

(U) All model parameters are derived from the following open-source references:

Table 23: (U) Sources

UNCLASSIFIED		
#	Reference	Content Used
[1]	DoDI 5000.61, Sep 2024	VV&A policy and documentation requirements
[2]	MIL-STD-3022, Jan 2008	VV&A documentation templates
[3]	Army Directive 2024-02	Software development policy
[4]	AAIB South Scarle Accident Report, 2023	T-150 specifications: MTOW, thrust, configuration
[5]	US Patent US20240262499A1, 2024	Battery topology: 13S, capacity, voltage
[6]	T-Motor U15II KV100 Specifications	Motor parameters: KV, max power, dimensions
[7]	AFSIM 2.2505 Documentation	Plugin API, mover/processor architecture
[8]	Cheeseman & Bennett, ARC R&M 3021, 1955	Ground effect model
[9]	Johnson, Helicopter Theory, 1980	Momentum theory, rotor aerodynamics
[10]	Leishman, Principles of Helicopter Aerodynamics, 2006	VRS theory and boundaries
[11]	SURVICE TRV-150 Product Data Sheet, 2024	Manufacturer specifications
		UNCLASSIFIED

7.2 (U) Overall Assumptions

1. (U) The T-150 operates in standard atmosphere (ISA) at sea level unless altitude effects are specified
2. (U) All 8 motors are identical and operate symmetrically
3. (U) The airframe is rigid (no structural deformation under load)
4. (U) No wind, turbulence, or atmospheric disturbances are modeled
5. (U) Sensor and navigation systems provide perfect state knowledge
6. (U) Battery cells are uniform and exhibit identical characteristics
7. (U) Payload is rigidly attached (except when using slung load processor)

7.3 (U) Overall Limitations

1. (U) **Flight Endurance Gap**: Simulated endurance (14.3 min) is approximately 40% of manufacturer specification (36 min). This is attributed to conservative power modeling and high auxiliary power estimates. Analysts should apply appropriate scaling when using endurance results.
2. (U) **No Environmental Effects**: Wind, temperature, humidity, and altitude effects on air density are not modeled. Results represent calm, sea-level conditions.
3. (U) **No Sensor Noise**: All state measurements are perfect. Real-world navigation, attitude estimation, and control will introduce errors not captured in simulation.
4. (U) **Informational VRS**: VRS detection is informational; it does not feed back to modify flight dynamics. In reality, VRS causes significant and potentially unrecoverable loss of lift.
5. (U) **Informational Slung Load Forces**: Slung load reaction forces are computed but do not modify aircraft motion. In reality, a swinging load significantly affects aircraft stability.

7.4 (U) Overall Gaps

1. (U) **No flight test data**: No actual T-150 flight test data is available for model validation. All validation is against manufacturer specifications, first-principles physics, and analytical predictions.
2. (U) **Inertia values**: Moments of inertia are estimated from platform geometry and mass distribution, not measured.
3. (U) **Battery chemistry specifics**: OCV curve and internal resistance characteristics use generic Li-Ion parameters, not the specific cell chemistry used in the T-150.
4. (U) **Motor-specific data**: Thermal mass, cooling rate, and efficiency curves are based on the T-Motor U15II class rather than specific test data.

8 (U) References

1. (U) Office of the Under Secretary of Defense for Research and Engineering, "DoD Instruction 5000.61: DoD Modeling and Simulation Verification, Validation, and Accreditation," September 17, 2024.
2. (U) Department of Defense, "MIL-STD-3022: Documentation of Verification, Validation, and Accreditation (VV&A) for Models and Simulations," January 28, 2008.
3. (U) Department of the Army, "Army Directive 2024-02: Software Development," 2024.
4. (U) Air Accidents Investigation Branch, "AAIB Investigation to Malloy Aeronautics T-150, South Scarle, Nottinghamshire," 2023.
5. (U) Malloy Aeronautics, "Battery Pack Topology for Unmanned Aerial Vehicle," US Patent US20240262499A1, 2024.
6. (U) T-Motor, "U15II KV100 Motor Specifications," 2024.
7. (U) Air Force Research Laboratory, "AFSIM 2.2505 Documentation: Advanced Framework for Simulation, Integration, and Modeling," 2025.
8. (U) I.C. Cheeseman and W.E. Bennett, "The Effect of Ground on a Helicopter Rotor in Forward Flight," ARC R&M 3021, 1955.
9. (U) W. Johnson, "Helicopter Theory," Princeton University Press, 1980.
10. (U) J.G. Leishman, "Principles of Helicopter Aerodynamics," 2nd ed., Cambridge University Press, 2006.
11. (U) SURVICE Engineering / Malloy Aeronautics, "TRV-150 Heavy-Lift UAS Product Data Sheet," 2024.

9 (U) Acronyms

Citations